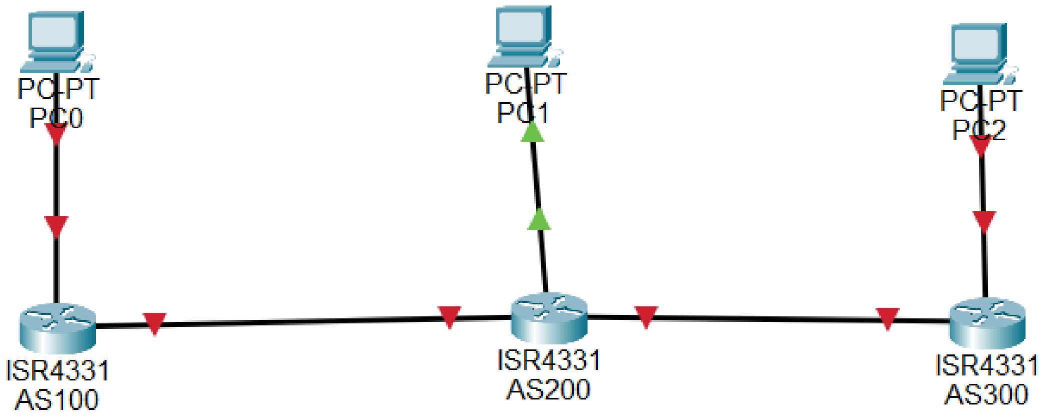


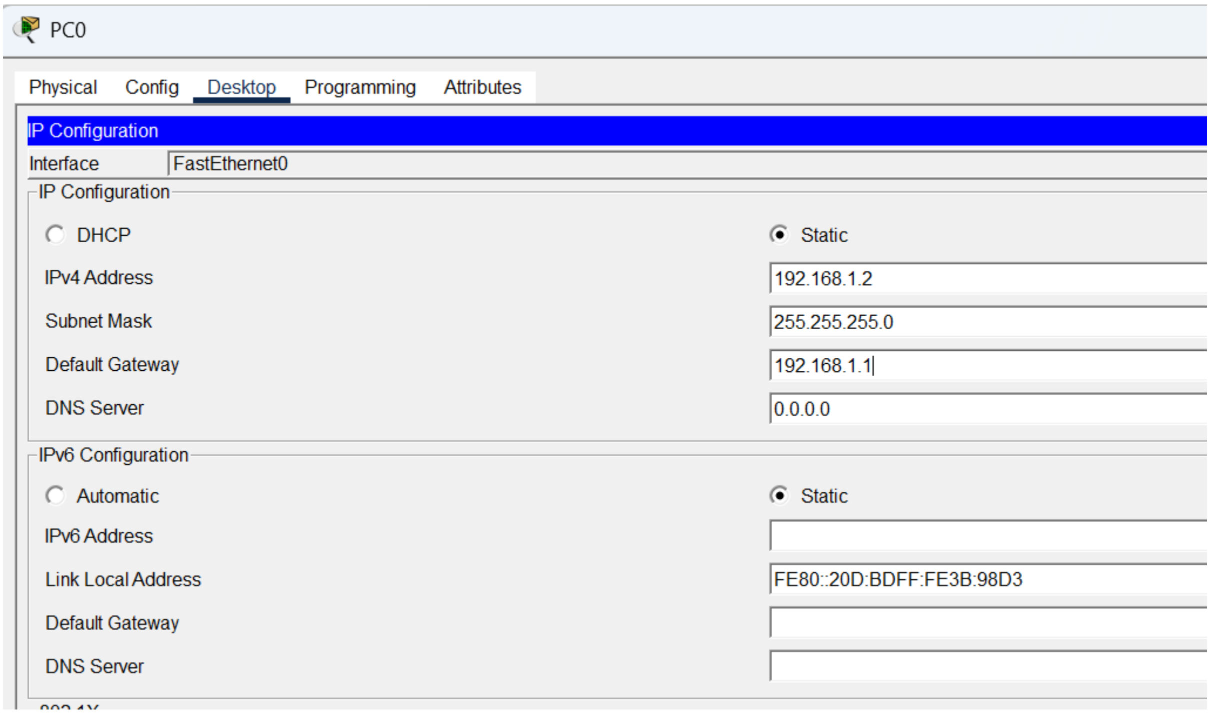

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#hostname R1
R1(config)#
R1(config)#interface g0/0/1
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
R1(config-if)#interface g0/0/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
R1(config-if)#router bgp 100
R1(config-router)#neighbor 10.0.0.2 remote-as 200
R1(config-router)#network 192.168.1.0 mask 255.255.255.0
R1(config-router)#
R1(config-router)#end
R1#wr
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R1#
```

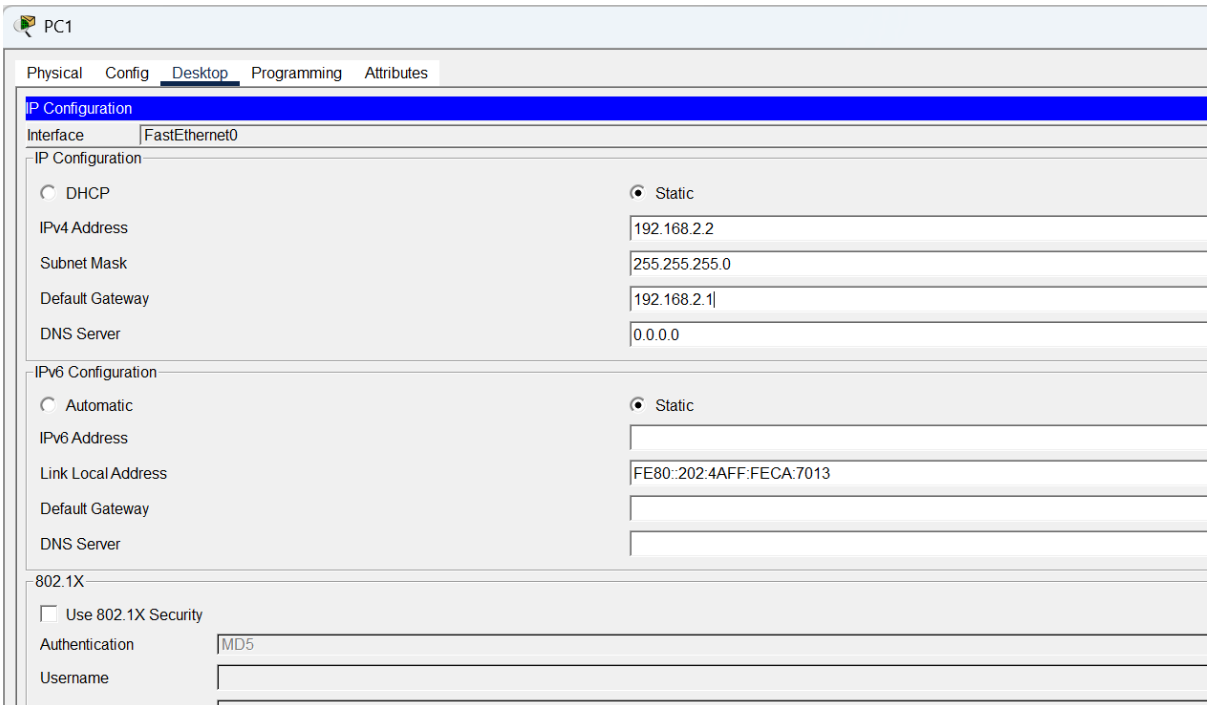
```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#hostname R2
R2(config)#
R2(config)#interface g0/0/1
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
R2(config-if)#interface g0/0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
R2(config-if)#interface g0/0/1
R2(config-if)#ip address 10.0.0.5 255.255.255.252
R2(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R2(config-if)#
```

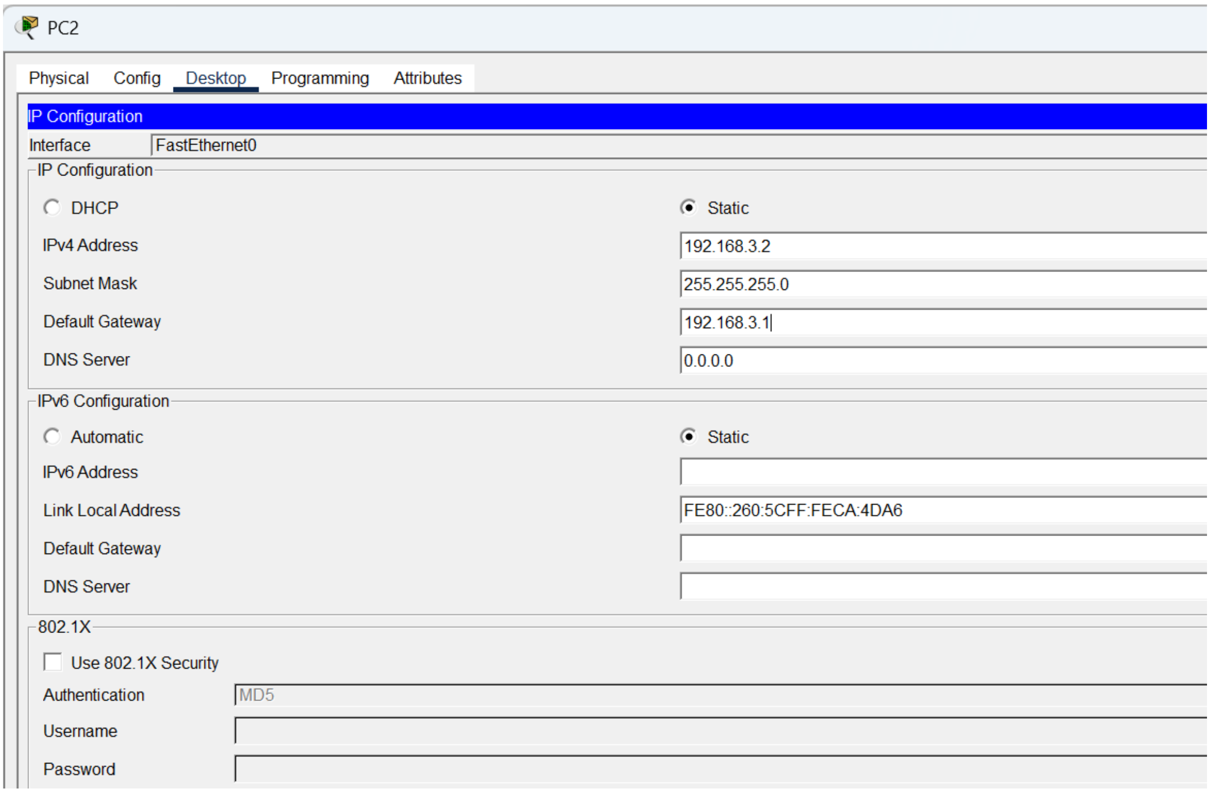
```
R2(config-if)#router bgp 200
R2(config-router)#neighbor 10.0.0.1 remote-as 100
R2(config-router)#neighbor 10.0.0.6 remote-as 300
R2(config-router)#network 192.168.2.0 mask 255.255.255.0
R2(config-router)#
R2(config-router)#end
R2#wr
%SYS-5-CONFIG_I: Configured from console by console
%BGP-5-ADJCHANGE: neighbor 10.0.0.1 Up

Building configuration...
[OK]
R2#
```

BGP for R2

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#hostname R3
R3(config)#
R3(config)#interface g0/0/1
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#no shutdown

R3(config-if)#
R3(config-if)#interface g0/0/0
R3(config-if)#ip address 10.0.0.6 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
R3(config-if)#router bgp 300
R3(config-router)#neighbor 10.0.0.5 remote-as 200
R3(config-router)#network 192.168.3.0 mask 255.255.255.0
R3(config-router)#
R3(config-router)#end
R3#wr
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
%BGP-5-ADJCHANGE: neighbor 10.0.0.5 Up

Building configuration...
[OK]
R3#
```




PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:4AFF:FECA:7013
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.3.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.3.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:5CFF:FECA:4DA6
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

```
[OK]
R1#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%BGP-5-ADJCHANGE: neighbor 10.0.0.2 Up

R1#show ip bgp summary
BGP router identifier 192.168.1.1, local AS number 100
BGP table version is 3, main routing table version 6
2 network entries using 264 bytes of memory
2 path entries using 104 bytes of memory
1/1 BGP path/bestpath attribute entries using 184 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 656 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.2        4   200       7       6        3    0    0 00:04:31        4

R1#
```

```
R2#show ip bgp summary
BGP router identifier 192.168.2.1, local AS number 200
BGP table version is 3, main routing table version 6
2 network entries using 264 bytes of memory
2 path entries using 104 bytes of memory
2/2 BGP path/bestpath attribute entries using 368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 816 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.1        4   100       7       6        3    0    0 00:04:59        4
10.0.0.6        4   300       6       5        3    0    0 00:03:56        4

R2#
```

```
[OK]
R3#show ip bgp summary
BGP router identifier 192.168.3.1, local AS number 300
BGP table version is 3, main routing table version 6
2 network entries using 264 bytes of memory
2 path entries using 104 bytes of memory
1/1 BGP path/bestpath attribute entries using 184 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 656 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.5        4   200       7       6        3    0    0 00:04:17        4

R3#
```

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time=15ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>
```

Check from PC0